

# Doping Stability Report -- Ni in LiCoO2 at 320 C

Generated: 2026-05-31 07:26 Host: LiCoO2 Dopant: Ni

## Executive Summary

Ni preferentially occupies the Co layer. The formation energy difference between the two layers is +8.187 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00313 Å<sup>2</sup>/ps (stable), compared to 0.00058 Å<sup>2</sup>/ps on the Li site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The Li-site E<sub>f</sub> is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

| Site | E <sub>f</sub> (eV) | Charge | Compensation | MSD pass | Coord pass | Verdict             |
|------|---------------------|--------|--------------|----------|------------|---------------------|
| Co   | +0.613              | +0     | No           | PASS     | FAIL       | STRUCTURAL COLLAPSE |
| Li   | +8.800              | +2     | Yes          | PASS     | FAIL       | STRUCTURAL COLLAPSE |

## Ni at the Co layer -- STRUCTURAL COLLAPSE

### Charge situation

Ni replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

### Formation Energy (thermodynamic stability)

|                                 |                   |             |                  |
|---------------------------------|-------------------|-------------|------------------|
| Formation energy E <sub>f</sub> | <b>+0.6128 eV</b> | <b>PASS</b> | < 1.0 eV to pass |
|---------------------------------|-------------------|-------------|------------------|

E<sub>f</sub> = +0.613 eV. This value is moderately positive, indicating that Ni incorporation is possible but endothermic; may require non-equilibrium synthesis at the Co site.

### Molecular Dynamics Stability (320 C, 25 ps)

|                                  |                                  |             |                                    |
|----------------------------------|----------------------------------|-------------|------------------------------------|
| MSD slope (late-time)            | <b>-0.00313 Å<sup>2</sup>/ps</b> | <b>PASS</b> | < 0.005 to pass (plateau = stable) |
| Final MSD                        | <b>0.055 Å<sup>2</sup></b>       | ----        |                                    |
| Max displacement                 | <b>0.599 Å</b>                   | ----        |                                    |
| Coordination (min/mean/max)      | <b>6 / 6.2 / 8</b>               | <b>FAIL</b> | relaxed = 6, tolerance +/-1        |
| Mean bond length (MD vs relaxed) | <b>1.996 Å (relaxed: 1.9)</b>    | ----        |                                    |
| Lattice volume change            | <b>+0.00 %</b>                   | <b>PASS</b> | < +/-5% to pass                    |

Late-time MSD slope = -0.00313 Å<sup>2</sup>/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.055 Å<sup>2</sup>; maximum displacement from starting position = 0.60 Å.

Coordination fluctuated between 6 and 8 (mean 6.2), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination\_cutoff\_A setting. Mean nearest-neighbour distance during MD = 1.996 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

**VERDICT: STRUCTURAL COLLAPSE**

### Doping Stability Report -- Ni in LiCoO<sub>2</sub> at 320 C

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ni atom barely moves (MSD = 0.055 Å<sup>2</sup>) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination\_cutoff\_A and re-running analysis.

## Ni at the Li layer -- STRUCTURAL COLLAPSE

### Charge situation

Ni replaces Li with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

### Formation Energy (thermodynamic stability)

|                                 |            |      |                  |
|---------------------------------|------------|------|------------------|
| Formation energy E <sub>f</sub> | +8.7998 eV | FAIL | < 1.0 eV to pass |
|---------------------------------|------------|------|------------------|

E<sub>f</sub> = +8.800 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Ni incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

### Molecular Dynamics Stability (320 C, 25 ps)

|                                  |                            |      |                                    |
|----------------------------------|----------------------------|------|------------------------------------|
| MSD slope (late-time)            | 0.00058 Å <sup>2</sup> /ps | PASS | < 0.005 to pass (plateau = stable) |
| Final MSD                        | 0.065 Å <sup>2</sup>       | ---- |                                    |
| Max displacement                 | 1.041 Å                    | ---- |                                    |
| Coordination (min/mean/max)      | 5 / 6.2 / 8                | FAIL | relaxed = 6, tolerance +/-1        |
| Mean bond length (MD vs relaxed) | 2.090 Å (relaxed: 2.0)     | ---- |                                    |
| Lattice volume change            | +0.00 %                    | PASS | < +/-5% to pass                    |

Late-time MSD slope = 0.00058 Å<sup>2</sup>/ps. The slope is near zero, confirming the dopant is tightly confined to its site. Final MSD = 0.065 Å<sup>2</sup>; maximum displacement from starting position = 1.04 Å.

Coordination fluctuated between 5 and 8 (mean 6.2), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination\_cutoff\_A setting. Mean nearest-neighbour distance during MD = 2.090 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

## VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ni atom barely moves (MSD = 0.065 Å<sup>2</sup>) but the coordination count fluctuates because the bond length at the Li site is close to the analysis cutoff. Consider increasing coordination\_cutoff\_A and re-running analysis.

## Site Preference Comparison

The Co site is preferred by 8.187 eV over the Li site. Formation energy: +0.613 eV (Co) vs +8.800 eV (Li).

Ni preferentially occupies the Co layer. The formation energy difference between the two layers is +8.187 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau

---

### Doping Stability Report -- Ni in LiCoO<sub>2</sub> at 320 C

slope is -0.00313 Å<sup>2</sup>/ps (stable), compared to 0.00058 Å<sup>2</sup>/ps on the Li site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The Li-site  $E_f$  is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

---

## Caveats and Limitations

### ? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch  $\neq 0$ ) are approximate even when structural compensation defects are included.

### ? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality  $E_f$  of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

### ? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all  $E_f$  values by additive constants; the relative ordering of sites is unaffected.

### ? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

### ? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.